

Richard Franks

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PROFESSIONAL SUMMARY

Principal Software Engineer with 15+ years of experience architecting and delivering large-scale software platforms across healthcare, fintech, and scientific domains. Proven track record leading the design of production AI systems, distributed web applications, enterprise frameworks, and cloud-native services used in mission-critical environments. Deep expertise in Python, Java, TypeScript, React, Next.js, Spring Boot, REST APIs, SQL/NoSQL data platforms, and modern AI technologies including retrieval-augmented generation (RAG), multi-LLM orchestration, evaluation pipelines, and medical transcription workflows. Recognized for rapidly mastering complex domains, driving architectural direction, mentoring engineers, and transforming difficult business workflows into scalable, maintainable software.

CORE SKILLS

Languages: Python, Java, JavaScript, TypeScript

Frontend & Full-Stack Frameworks: React, Next.js, NestJS, Spring, Spring Boot, Django, Flask, Hibernate

Backend & Integration: REST APIs, web services, backend services, service integration, full-stack development, monorepo development

Databases & Data Modeling: SQL, MySQL, PostgreSQL, Oracle, CosmosDB, MongoDB, relational and non-relational data modeling

AI & Applied ML: LLMs, RAG, LangChain, AI workflow orchestration, model evaluation, transcription pipelines (STT and TTS), Claude, Gemini, OpenAI, Cursor

Cloud & Platform: Azure Functions, AWS, cloud-native application development, third-party API integration

Delivery & Quality: CI/CD, Docker, Containerization, GitHub Actions, Jenkins, automated testing, testing automation, Jest

Leadership & Execution: Architecture, code reviews, technical mentoring, cross-functional collaboration, end-to-end product development

EDUCATION

University of Southern Maine, 2008 — M.S., Computer Science

Boston University, 2003 — B.S., Biomedical Engineering

EXPERIENCE

Carebrain — Distinguished Engineer

Remote | Sep 2025 – Mar 2026

- Architected a production AI platform powering clinical documentation, intelligent routing, and healthcare workflow automation using TypeScript, Next.js, NestJS, LangChain, and multiple commercial LLM providers.
- Designed a vendor-agnostic LLM orchestration framework capable of dynamically selecting models, evaluating response quality, and automatically failing over between providers to improve reliability while eliminating vendor lock-in.
- Led development of an in-person clinical scribe pipeline integrating Deepgram, AssemblyAI, AWS Transcribe Medical, AWS Comprehend Medical, and ElevenLabs for transcription, diarization, entity extraction, and clinical insight generation.
- Engineered HIPAA-compliant auditing across every PHI transmission path, including internal APIs, databases, third-party AI providers, and external healthcare services.
- Built clinician-facing patient intelligence tools capable of combining traditional application logic with AI-generated insights derived from structured and unstructured EHR data.
- Improved transcription quality and clinical workflow accuracy through multi-model consensus strategies and automated evaluation pipelines.
- Developed a real-time clinician collaboration platform using GetStream, enabling Slack-style secure clinical communication.

PatientKeeper / HCA / Commure — Principal Software Engineer

Waltham, MA | Remote | Aug 2012 – Sep 2025

- Architected and delivered production AI capabilities over enterprise electronic health record data, combining retrieval-augmented generation, vector search, multiple LLM providers, and clinical knowledge to support summarization, question answering, and workflow automation.
- Extended AI capabilities to identify complications, risk factors, diagnoses, and coding opportunities directly from physician documentation using large language models and structured clinical data.
- Modernized the company's enterprise portal architecture by designing a modular framework supporting hundreds of independently deployable clinical applications through a REST-based service architecture.
- Helped build one of the organization's core dashboard frameworks supporting more than 200 reusable clinical widgets across multiple healthcare product suites.
- Designed reusable enterprise UI components adopted throughout the engineering organization, including patient selection, patient assignment, and patient list management frameworks.
- Served on the Architecture Board, influencing platform direction, reviewing major technical initiatives, mentoring engineers, and establishing engineering standards across teams.
- Led cross-functional architecture discussions spanning frontend, backend, infrastructure, AI services, and clinical product development.
- Championed software quality through design reviews, technical documentation, automated testing, CI/CD improvements, and engineering mentorship.

DressageMarket.com, LLC — Chief Engineer / Entrepreneur

Remote | Aug 2009 – Mar 2021

- Co-founded and served as sole software engineer for a web-based equestrian marketplace, owning product strategy, software architecture, implementation, deployment, and long-term platform evolution.
- Designed and developed the platform using Python and Django, delivering a complete end-to-end marketplace supporting buyers, sellers, and business partners.
- Worked directly with co-founders to translate product vision into technical strategy, rapidly delivering new features while maintaining a stable production environment.
- Managed every aspect of the software lifecycle, including architecture, database design, backend services, frontend development, deployment, production support, and continuous enhancement.

REZ-1 — Software Engineering Consultant

Wellesley, MA | Oct 2010 – Jun 2012

- Contributed across the full technology stack during a large-scale modernization of enterprise freight marketplace software used throughout the transportation industry.
- Developed user interfaces, backend services, persistence layers, automated tests, deployment tooling, and framework enhancements using Java, Spring, Hibernate, Vaadin, Liferay, Tomcat, JPA, and JAX-WS.
- Helped design annotation-driven development frameworks that accelerated feature delivery while reducing repetitive implementation effort across engineering teams.
- Collaborated closely with architects and senior engineers on large-scale enterprise application design and modernization initiatives.

CashStar — Engineer

Portland, ME | Jun 2009 – Oct 2010